

LAHARI R

Bachelor of Engineering in CSE (Data Science) Engineering

ATME College of Engineering, Mysore

+91 7411479659 | lahari22004@gmail.com | www.linkedin.com/in/lahari-r-71b6ab302

SUMMARY

Computer Science graduate with hands-on experience in Python backend development, FastAPI, REST APIs, PostgreSQL, and cloud deployment. Built scalable backend applications, API services, and AI-powered systems using FastAPI, Docker, AWS, and Git. Strong understanding of databases, backend architecture, and API development with a passion for building reliable and efficient software systems.

EDUCATION

- | | |
|---|------------------|
| • ATME College of Engineering, B.E in Computer Science – Data Science | CGPA 8.71 |
| • Gopalswamy PU College, Pre-University | Percentage 85% |
| • Christ Public School, 10 th | Percentage 75% |

SKILLS

- **Backend Development:** FastAPI, Flask, REST APIs, API Integration, Authentication & Authorisation
- **Programming Languages:** Python, SQL, C
- **Python Libraries:** Pandas, NumPy, Matplotlib, TensorFlow, OpenCV, Pydantic
- **Databases:** MySQL, MongoDB, SQL, PostgreSQL, NoSQL Basics
- **Cloud & DevOps:** Docker, AWS EC2, S3, RDS, GitHub Actions
- **Tools:** Git, GitHub, Postman, Linux
- **AI/ML:** Machine Learning, CNN, LSTM, Data Preprocessing, Feature Engineering

PROFESSIONAL EXPERIENCE

AI & ML Intern | Orvionar Private Ltd (Feb 2026 – May 2026)

Completed an offline internship focused on Machine Learning and Artificial Intelligence, working on model development, data preprocessing, and real-world AI applications. Gained hands-on experience with Python, TensorFlow, data analysis, and problem-solving techniques.

Cloud Engineering & AWS Intern | Eighth Note Internet Pvt. Ltd. (Feb 2026 – May 2026) (Online)

Worked on cloud engineering and AWS services, including deployment, infrastructure basics, and automation concepts. Gained practical exposure to AWS services (EC2, S3, Lambda, RDS), DevOps fundamentals, and scalable application deployment.

PROJECTS

Multi-Document RAG Chatbot using LangChain, FastAPI & PostgreSQL

Developed a Retrieval-Augmented Generation (RAG) chatbot capable of answering questions from multiple PDF documents by building REST APIs using FastAPI. Implemented document processing, text chunking, embedding generation, and semantic search using LangChain and PostgreSQL-based vector storage. Designed scalable backend services for document indexing and retrieval, integrated Large Language Models for context-aware responses, and containerized the application using Docker while managing version control with Git.

Key Skills: Python, FastAPI, LangChain, RAG, PostgreSQL, Vector Databases, REST APIs, LLM Integration, Docker, Git

Simplicode – Multi-Language Cloud Code Runner

Developed a browser-based cloud code execution platform that enables users to write and execute Python, C, and Java programs in real time. Built backend services using Flask for code execution, process management, timeout handling, and temporary file cleanup. Integrated Monaco Editor to provide a VS Code-like experience, designed REST-based functionality for code execution, and deployed the application on the cloud using Render with Git-based version control.

Key Skills: Python, Flask, Backend Development, REST APIs, Process Management, Error Handling, Web Development, Docker, Git, Cloud Deployment

GitHub: <https://github.com/laharirr/Simplicode>

Live Link: <https://simplicode.onrender.com>

DeployML – Cloud-Based ML Deployment Platform

Built a cloud-based machine learning deployment platform using FastAPI to provide real-time prediction APIs for machine learning models. Containerized the application using Docker and deployed backend services on AWS EC2. Implemented CI/CD pipelines using GitHub Actions for automated build and deployment processes while designing scalable API endpoints for model inference and cloud-based application deployment.

Key Skills: Python, FastAPI, Docker, AWS EC2, CI/CD, GitHub Actions, API Development, Cloud Deployment, Machine Learning

GitHub: <https://github.com/laharirr/DeployML>

Live Link: <http://16.171.231.71:8000/docs>

AI-Powered Energy Consumption Forecasting and Prediction

Developed an intelligent energy forecasting system using Python and time-series models including LSTM, ARIMA, and Prophet to predict energy consumption patterns. Performed data preprocessing, feature engineering, and model optimization to improve prediction accuracy. Built an interactive Streamlit dashboard for real-time visualization and analysis, enabling users to monitor and interpret forecasting results effectively.

Key Skills: Python, LSTM, ARIMA, Prophet, Time-Series Forecasting, Data Preprocessing, Feature Engineering, Streamlit, Machine Learning

Sign Language Detection

Designed and developed a real-time hand gesture recognition system using Python, TensorFlow, OpenCV, and MediaPipe for sign language prediction. Collected and processed gesture datasets, trained a CNN-based deep learning model, and implemented live webcam-based gesture detection. Automated code management and deployment workflows using Git and GitHub Actions to ensure efficient version control and project maintenance.

Key Skills: Python, TensorFlow, CNN, OpenCV, MediaPipe, Computer Vision, Deep Learning, Image Processing, GitHub Actions

GitHub: <https://github.com/laharir/sign-language-detection>

CERTIFICATIONS

- Data Science for Engineers– NPTEL.
- Machine Learning Foundation Certification (Infosys Springboard)
- Explore machine learning using Python (Infosys Springboard)
- Introduction to Data Science (CISCO)
- Data Analytics Essentials (CISCO)
- Domain Specific Training on Applied Machine Learning (IVIS LABS)

PUBLICATIONS

“Survey on AI-Powered Energy Consumption Forecasting and Prediction” – Published as part of academic research at *ATME College of Engineering, Mysuru (2025)*.

STRENGTHS

- Strong proficiency in Python, FastAPI, and REST API development.
- Good understanding of databases, backend architecture, and API integration.
- Hands-on experience with AWS, Docker, Git, and cloud deployment.
- Strong problem-solving, debugging, and analytical skills.
- Quick learner with the ability to adapt to new technologies and frameworks.